Kevin Thomas Luzbetak

Senior Data Engineer • Senior Software Engineer • Machine Learning Engineer

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📍 Agoura Hills, CA | U.S. Citizen | 25 years in software & data systems

Summary

Cloud Databases • Distributed Systems
I bring 25+ years of experience in software and data engineering, including architecting and scaling distributed data platforms using Apache Spark, Databricks, Snowflake, and Amazon Redshift across AWS and Azure. Experienced in performance optimization, cloud migrations, and building ETL and analytics pipelines leveraging Medallion Architecture.

Machine Learning (AI)
Developing machine-learning and AI-driven applications, retrieval-augmented generation (RAG). Hands-on experience with LangChain Community, Ollama, Huggingface, Databricks IQ, Snowflake Cortex AI along with full-stack development across both frontend Node.js and backend systems.

Data Engineering • Data Modeling
Building ETL data pipelines and large-scale analytics solutions on Databricks Delta Lake, Snowflake, and Redshift. Experienced in physical, hierarchical, relational, dimensional, and canonical data modeling, enabling optimized storage, analytics, and integration across complex systems.

Education

Master's Degree in Computer Science • *Graduated 2014*
California Lutheran University • Thousand Oaks, California
Focus: Machine Learning, Distributed Systems, Computer Vision, Embedded Systems

Bachelor's Degree in Information Technology • *Graduated 2004*
University of Phoenix • Woodland Hills, California
Focus: Software Development, Database Systems, Artificial Intelligence

Certificate in C/Linux • *Graduated 2001*
UCLA Extension • Los Angeles, California

Technical Skills

Programming Languages:
Python, PySpark, SQL, Scala/UDF, Bash, C++, GoLang, Java, Node.js

AI & Data Science:
Machine Learning, Retrieval-Augmented Generation (RAG), FAISS Retrieval QA, Databricks MLflow, Snowflake Cortex AI, Natural Language Processing (NLP), PyTorch, TensorFlow, MLlib, Pandas, NumPy, Data Visualization, Tableau, Telemetry

Data Platforms & Databases:
Databricks Delta Lake, Snowflake Keystone, Apache Iceberg, Apache Spark, Unity Catalog, Medallion Architecture, Hadoop, Kafka, MySQL, PostgreSQL, Cassandra

Data Modeling:
Conceptual, Logical, and Physical Modeling; Kimball Dimensional Modeling; Star and Snowflake Schemas; Canonical Data Models; Entity–Relationship (ER) Modeling; Data Lineage and Metadata Modeling; Master Data Management (MDM); Semantic Layer Design

Infrastructure, DevOps & Monitoring:
Ubuntu Admin, Networking, GitHub, Apache Airflow, Jenkins, Terraform, Docker, Grafana, Splunk, Nagios

Professional Experience

Senior Data Engineer – Databricks Inc. (Contract)
San Francisco, CA · Dec/2024 – Apr/2025
Programmed enterprise HR data platform and API integration framework for People Systems, architecting automated solutions processing recruitment and employee lifecycle data.

- **Programmed API integration framework** in Python for Greenhouse and Workday systems, implementing OAuth authentication, retry mechanisms, rate limiting, and error handling for reliable extraction of 100K+ employee records and recruitment data
- **Architected database schema** using Medallion Architecture on Delta Lake, Unity Catalog, designing normalized data models for employee lifecycle, recruitment funnel, and organizational hierarchy with SCD Type 2 for historical tracking
- **Programmed custom anomaly detection algorithms** using PySpark and statistical methods, identifying data quality issues in candidate applications, interview feedback, and employee records with 95% accuracy
- **Programmed automated data cleansing pipelines** with Python, implementing regex patterns, fuzzy matching, and business rule engines to standardize and validate HR data, reducing manual data correction.
- **Programmed CLI tools and utilities** for database operations, building command-line interfaces for schema deployment, data validation, bulk updates, and report generation used by 50+ internal stakeholders

- **Programmed real-time analytics platform** processing streaming updates from HR systems, enabling live dashboards for hiring metrics, employee attrition, and diversity analytics
- **Programmed comprehensive testing framework** with pytest and Great Expectations, implementing unit tests for API integrations and data quality validation reducing production issues.

Technologies: Python, PySpark, SQL, Databricks Delta Lake, REST APIs, AWS (S3, Lambda), CLI development, Git, OAuth, Medallion Architecture, Unity Catalog

Senior Data Engineer – Disney Streaming (Contract)
Santa Monica, CA · May/2023 – Mar/2024
Programed Lifecycle Marketing Analytics to support lifecycle marketing analytics and data science use cases, including engagement experiences, audience insights, campaign performance, and automating and scaling customer data science model workflows.

- **Developed domain-oriented ETL frameworks** with Python/PySpark, creating reusable libraries and abstract base classes for data ingestion, transformation, and validation.
- **Built custom Data Dictionary** using full-text search engine for database objects table names, fields name and stored procedures, and enhanced Unity Catalog.
- **Programmed AI-powered SQL generation system** leveraging Snowflake Cortex, developing natural language interfaces that automatically generated optimized queries, reducing analyst query development time.
- **Engineered cross-platform data synchronization between Snowflake and Databricks** using Keystone/Iceberg table format, implementing ACID transactions, time-travel capabilities, and schema evolution for petabyte-scale datasets
- **Developed performance optimization framework implementing partition pruning,** Z-ordering, and adaptive query execution, achieving 5x improvement in analytical query performance while reducing compute costs.
- **Built CI/CD infrastructure using Terraform, Airflow, and Jenkins,** developing automated testing suites, deployment pipelines, and rollback mechanisms for database schema changes and data pipeline updates.

Technologies: Python, PySpark, Snowflake, Databricks, Apache Iceberg, Keystone, Cortex AI, AWS, Terraform, Airflow, Jenkins, Collibra, SQL, Unity Catalog

Senior Data Engineer – Nike Inc. (Contract)
Beaverton, OR · Feb/2022 – May/2023
Programed enterprise data platform serving Nike's $50B+ global operations, focusing on scalable software solutions and database architecture design.

- **Programmed custom ETL framework** in Python/PySpark with reusable components, abstract classes, and design patterns (Factory, Strategy, Observer) for pipeline orchestration, reducing development time for new data flows.
- Designed Medallion database architecture implementing (Bronze/Silver/Gold layers) on Delta Lake, establishing ACID-compliant transaction processing for 1B+ daily events with optimized partition strategies and Z-ordering for sub-second query performance

- **Programmed software infrastructure for Snowflake-to-Databricks migration**, writing custom connectors, parallel processing algorithms, and automated schema evolution tools handling 300+ TB with zero-downtime cutover
- **Programmed data catalog application** using Python Flask/FastAPI backend with full-text search algorithms, metadata parsing, and lineage tracking across 10,000+ database objects
- **Programmed infrastructure-as-code using Terraform modules and Python SDK,** automating database provisioning, cluster configuration, and security policies across multi-region deployments
- **Programmed automated testing framework with Pytest**, implementing unit tests, integration tests, and data quality validation.
- **Optimized database performance** through custom indexing strategies, materialized views, and query optimization, achieving 4x improvement in analytical query response times

Technologies: Python, PySpark, Databricks, Delta Lake, Snowflake, AWS, Terraform, Airflow, Jenkins, SQL,Git

Databricks SRE Engineer – Brighthouse Financial (Contract)
Charlotte, NC · Aug/2021 – Jan/2022
Programed critical financial data infrastructure from on-premises to Azure, supporting $100B+ in assets under management.

- **Orchestrated enterprise-wide migration** from on-premises Hadoop to Azure Databricks, successfully transferring 500+ TB of financial data with zero downtime during market hours
- **Programmed automated ETL pipelines** using Azure Data Factory processing 50M+ daily transactions, reducing batch processing time from 8 hours to 2 hours for regulatory reporting
- **Optimized Spark job performance** through advanced tuning techniques, achieving 65% reduction in compute costs while improving query response times by 4x for actuarial models
- **Implemented time-series data architecture** on Azure Storage with automated backup strategies, ensuring 99.99% availability for historical financial data spanning 20+ years
- **Collaborated with Microsoft Azure engineering team** to resolve critical cluster stability issues, improving platform reliability from 95% to 99.9% uptime
- **Designed auto-scaling Ambari clusters** using Azure DevOps CI/CD pipelines, dynamically adjusting resources based on workload demands and saving $50K monthly in compute costs
- **Programmed monitoring and alerting framework for data quality validation**, catching anomalies before they impacted downstream financial reporting

Technologies: Azure (Databricks, Data Factory, Storage, DevOps), PySpark, Python, Ambari, Hadoop, SQL, Time-series databases, Prometheus

Senior Data Engineer – Apple Inc. (Contract)
Cupertino, CA · Jan/2021 – Jun/2021
Programed Apple Pay Wallet, processing millions of payment transactions daily across the global iOS ecosystem.

- **Programmed ETL pipeline** for Apple Pay credit card synchronization, ensuring sub-second latency updates across millions of user devices while maintaining PCI compliance
- **Implemented Apache Iceberg-based data lakehouse** on Spark, processing 10M+ daily transactions with ACID guarantees and time-travel capabilities for payment reconciliation
- **Programmed fault-tolerant Kafka streaming architecture** handling real-time payment events data consistency across distributed systems and Scala UDFs for secure tokenization and encryption of sensitive payment data, meeting Apple's stringent security requirements.
- **Programmed testing framework** code coverage, preventing production incidents in payment-critical systems handling billions in transaction volume
- **Optimized Spark workloads** reducing processing while improving pipeline throughput partition strategies.
- **Iceberg implementation** with Apple's broader open-table architecture initiative, establishing patterns for future payment services

Technologies: Apache Spark, Apache Iceberg, Apache Kafka, Scala, AWS (EMR, S3), PCI-compliant infrastructure

Database Engineer III – Amazon Web Services (AWS)
East Palo Alto, CA · Aug/ 2019 – Jan/2021
Programed data warehouse migrations for Fortune 500 clients, moving petabyte-scale workloads from legacy systems to AWS cloud infrastructure.

- **Executed 50+ production migrations** from Teradata/Netezza to Redshift using AWS DMS, successfully transferring 100+ TB datasets with zero data loss and minimal downtime windows
- **Programmed high-performance migration pipelines** achieving 10GB/s throughput, reducing migration timelines from months to weeks for multi-petabyte data warehouses
- **Programmed automated data validation framework** using Python, performing row-count reconciliation, column checksums, and CDC verification across billions of records, ensuring 100% data integrity
- **Programmed infrastructure-as-code automation using AWS CLI and Bash**, reducing migration setup time by 80% and enabling repeatable, self-healing migration patterns.
- **Optimized Redshift cluster performance** post-migration, implementing automated vacuum/ analyze schedules and WLM configurations that improved query performance.
- **Mentored client teams on AWS best practices**, enabling self-sufficient management of their migrated data platforms and reducing support tickets.
- **Created real-time monitoring dashboards with CloudWatch**, proactively identifying and resolving replication lag issues

Technologies: AWS DMS, Amazon Redshift, AWS CLI, Python, PySpark, Bash, CloudWatch, S3, IAM, Teradata, Netezza

Senior Software Engineer – Nexstar Media Group, Inc.
Playa Vista, CA · Apr/2018 – July/2019
Programed full-stack streaming solutions for major TV networks and national weather services, combining real-time content delivery with predictive analytics.

- **Programmed RESTful microservices** handling millions of API requests daily for TV station websites, achieving 99.95% uptime during breaking news and severe weather events
- **Built machine learning models analyzing** user engagement across streaming platforms, processing session logs and interaction data to predict viewer behavior with 85% accuracy
- **Developed real-time analytics pipeline** using PySpark and ElasticSearch, enabling TV stations to optimize content placement and increase viewer retention.
- **Designed PostgreSQL data warehouse** integrating subscription, viewership, and demographic data, enabling cross-platform attribution and ROI measurement for marketing campaigns
- **Created responsive web applications serving** weather alerts and breaking news to millions of viewers, with auto-scaling infrastructure handling 10x traffic spikes during major events

Technologies: AWS (EC2, Lambda, RDS), Python, PySpark, Java, PostgreSQL, ElasticSearch, Prometheus, React, Node.js

Senior Data Engineer – Disney Studios / Movies Anywhere (Contract)
Burbank, CA · July/2017 – Apr/2018
Programed data infrastructure processing millions of daily streaming events across major studio partnerships (Disney, Universal, Sony, Warner Bros, Fox).

- **Programmed distributed Spark streaming pipelines** on EMR clusters, processing 100M+ daily events with sub-second latency for real-time user behavior analytics
- **Programmed AWS data lake architecture using S3/Athena,** enabling self-service analytics for business teams across movie consumption patterns
- **Programmed geospatial analysis** and session reconstruction algorithms to track cross-platform viewing behavior, directly informing content licensing decisions worth millions
- **Programmed fault-tolerant ETL pipelines** with automated data quality monitoring, achieving 99.9% data accuracy for critical royalty payment calculations including Apple Music integration
- **Programmed serverless Lambda functions** for event-driven processing, reducing infrastructure costs by 40% while improving pipeline resilience and scalability

Technologies: AWS (EMR, S3, Athena, Lambda, Kinesis), PySpark, Python, Hadoop, SQL, Airflow, Kubernetes,

Solution Architect – DataStax, Inc.
Santa Clara, CA · Oct/2015 – Jul/2017
Programed Cassandra-Databricks integration solutions, enabling real-time analytics at scale for enterprise clients including Intuit.

- **Designed and deployed multi-datacenter Cassandra clusters** across AWS and Azure, implementing geo-distributed architectures that achieved sub-100ms latency for bi-coastal operations
- **Programmed high-throughput data pipelines** using PySpark and Scala UDFs, processing millions of transactions per second during Intuit's peak tax season with zero downtime
- **Led critical incident response in Intuit war room during tax season surges**, scaling clusters dynamically to handle 10x normal load while maintaining SLA performance targets
- **Delivered production-ready proof-of-concepts for Fortune 500 clients,** translating complex requirements into scalable solutions that reduced data processing costs by optimizing cluster utilization
- **Programmed automated performance monitoring and capacity planning systems**, proactively identifying bottlenecks and optimizing network topology across multiple availability zones

Technologies: Apache Cassandra, Databricks, AWS (EC2, S3, VPC), Azure, Python, PySpark, Scala, SQL

Senior Platform Engineer – MD Insider, Inc.
Santa Monica, CA · 11/2014 – 10/2015

- Developed a machine learning platform to analyze and aggregate billions of medical records, applying multi-class classification on structured patient data for churn prediction, medical diagnosis, and fraud detection using a Feedforward Neural Network.
- MD Insider platform delivered data-driven insights into physician performance, evaluating clinical experience, quality, and cost metrics. It supported healthcare organizations and employers in optimizing provider networks, enabling better decision-making, and ensuring transparency in matching patients with high-performing doctors based on comprehensive, risk-adjusted data.

Technologies: AWS (EMR, S3, EC2), Databricks/Spark, Scala, Python, TensorFlow/Scikit-learn, SQL

Staff Software Engineer – Skyworks Solutions, Inc.
Moorpark, CA Aug 2013 – Nov 2014

- I developed and maintained automated quality measurement systems for semiconductor manufacturing process control, focusing on RF chips used in Apple iPhones. I implemented a Six Sigma–driven analytics system to monitor manufacturing processes and ensure compliance with Process Capability (CPK) standards. Leveraging Six Sigma methodologies, particularly DMAIC (Define, Measure, Analyze, Improve, Control)
- I integrated data science and statistical process control (SPC) techniques into the production workflow. This enabled early detection of process variations, defect elimination, and assurance of high product quality through real-time data analysis. To support these efforts, I

built systems to collect and analyze production data, generating real-time product quality reports. Technologies used included Java, Oracle for data modeling, and data warehousing.

Technologies: Python, Java, Oracle, Unix/Linux, SPC/Six Sigma methodologies

Senior Software Engineer – AT&T Interactive Yellow Pages
Glendale, CA • 2010 – 2013
Programed Yellow Pages AI team classification business listings.

- **Programmed NLP-driven classification models to categorize business listings based on unstructured text data**, processing over 20 million business descriptions in automated categorization across 5,000+ business categories using techniques like TF-IDF vectorization and named entity recognition.
- **Programmed and analyzed large datasets to automate business-category assignment workflows**, reducing manual review time through data validation pipelines, anomaly detection algorithms.
- **Programmed decision-tree models for structured classification logic and workflow automation**, implementing C4.5 and CART algorithms to create interpretable rule-based systems that business analysts for faster category assignment for ambiguous listings.
- **Programmed model accuracy and generalization by applying Random Forest**, combining 100+ decision trees to reduce overfitting and improve prediction stability.
- **Programmed feedforward neural networks for feature extraction and category prediction**, designing multi-layer perceptrons with 3-5 hidden layers to automatically learn complex patterns from business descriptions, phone numbers, and location data.
- **Collaborated with the AI team to scale data processing pipelines using Hadoop and Linux environments**, implementing MapReduce jobs for distributed model training, optimizing data partitioning strategies, and maintaining cluster performance for 150+ node Hadoop infrastructure supporting real-time classification services.

Technologies: Python (NumPy, SciPy, NLTK), Java, Hadoop (MapReduce, HDFS), Linux (Shell scripting, Cron jobs), Machine Learning libraries, SQL databases.